

Comptia A+ Assignment

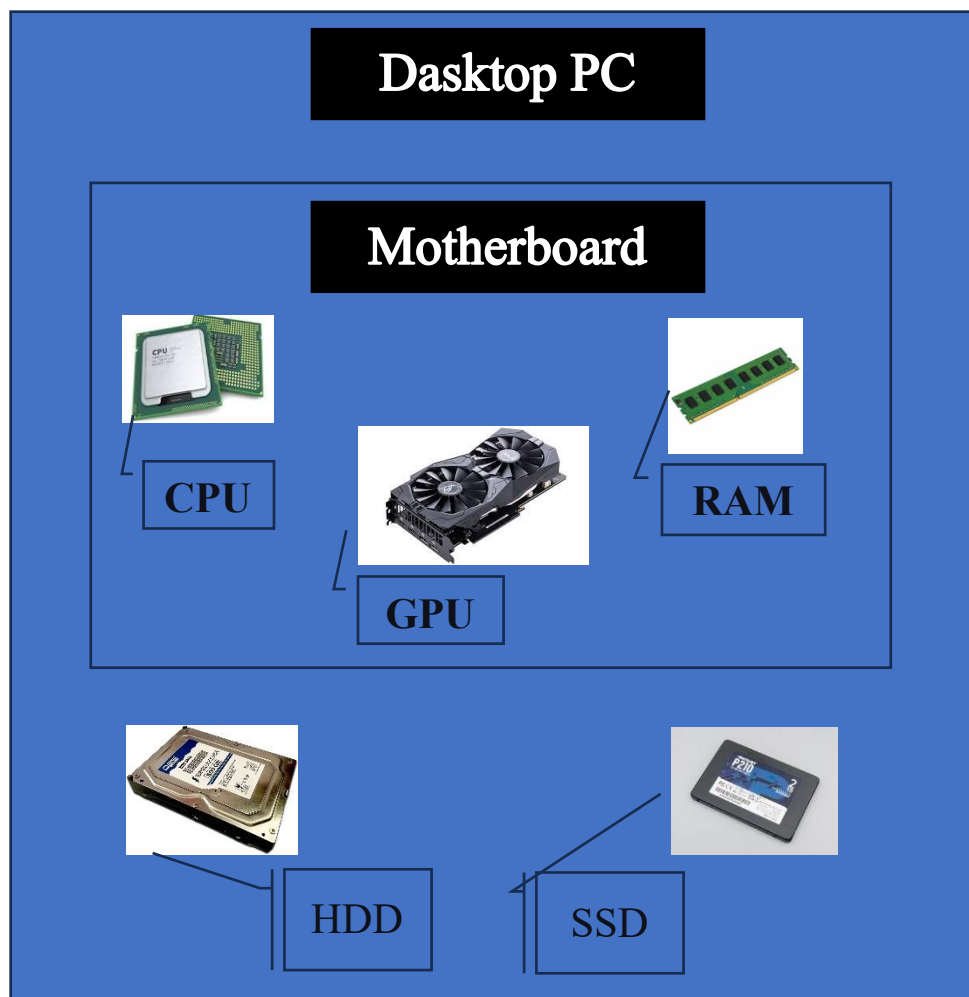
Module -1, 2 [Hardware and its components]

Overview of Hardware Components Theory.

Create a diagram of a PC with labeled hardware components.

Task

- ❖ 1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



- ❖ 2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.
 Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Storage	Uses spinning magnetic disk and a moving read/write head.	Uses flash memory chips with no moving parts.
Speed	HDD speed is slower. Takes more time to boot the system and open file.	SSD speed is faster. Boots the system quickly and loads application in seconds.
Durability	Less durable because it has moving mechanical parts. It can be damaged by shocks and drops.	More durable because it has no moving parts. It is highly resistant to shocks and vibration.
Performance	Slower file transfer and lower overall performance.	Faster file transfer and better overall performance.
Cost	Cheaper per GB of storage.	More expensive per GB of storage.
Typical Use Cases	Used in desktop pc, laptop and system where large storage is needed at a lower cost.	Used in modern laptop, gaming pc, high-performance computer.

❖ 3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

➤ Below describing the role of CPU, RAM, GPU

1. CPU

- The CPU is the brain of the computer.
- It processes game instruction, controls game logic and manages tasks such as player actions and communication between hardware components.
- A faster CPU provides smoother gameplay and better overall performance.

2. RAM

- RAM temporarily stores the game data and files that the computer is currently using.
- It allows the game to load quickly and switch between different game scenes without delays.
- More RAM helps reduce lag and improves multitasking while gaming.

3. GPU

- The GPU is responsible for rendering graphics, textures, lighting, and visual effects in game like PUBG and Free Fire.
- It ensures smooth frame rates and high-quality visuals, making the gaming experience more realistic and enjoyable.

- A powerful GPU allows games to run at higher resolutions and graphic setting.
- ❖ 4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.
 - so, to run the latest version of FIFA smoothly, I would upgrade the below hardware components.
 - ✓ **CPU: -**
The CPU handles game calculations, AI, and overall system performance. Upgrading to a faster processor reduces loading times and prevents performance bottlenecks during gameplay.
 - ✓ **GPU: -**
The Graphics Card is the most important component for gaming. A powerful GPU improves graphics quality, increases FPS (Frames Per Second), and provides a smooth gaming experience without lag.
 - ✓ **RAM: -**
Having at least 16 GB RAM allows the game and background applications to run smoothly. More RAM improves multitasking and reduces stuttering while gaming.
 - ✓ **SSD: -**

Replacing a Hard Disk Drive (HDD) with a Solid-State Drive (SSD) significantly reduces game loading times, speeds up Windows startup, and improves overall system responsiveness.

✓ **Power Supply Unit (PSU): -**

A more powerful graphics card and processor require additional power. Upgrading the PSU ensures stable power delivery and protects the system from power-related issues.